

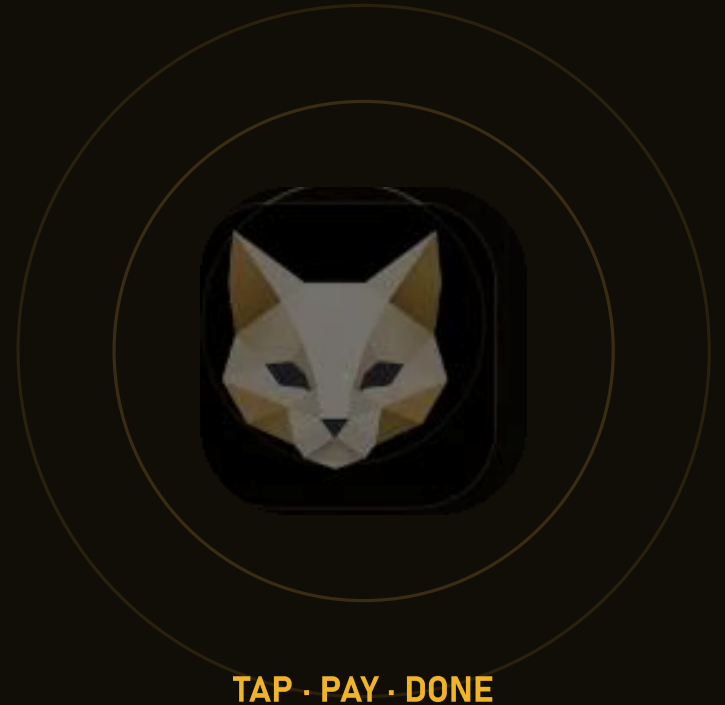
CONTACTLESS PAYMENTS ON STELLAR

Noir Wallet

Tap. Pay. Done.

A wallet you never open. Payments that just happen. Tap any RFID sticker, NFC card, or wearable to pay instantly — no app, no unlock, no confirmation.

Presented by: Richie Christian De Guzman



Paying Is Still Full of Friction

Contactless was supposed to be instant. In reality, every payment still demands attention — unlock the phone, open the app, authenticate, confirm. **Crypto is even harder to spend: you need a cold wallet or a phone app just to pay for anything.** Behind it all, merchants surrender percentage fees to card rails and wait days for settlement — and high-throughput venues can't afford either.

10–30s

Per Checkout Interaction

Unlock, open, authenticate, confirm — every single time

2–3%

Card Rail Fees

What merchants surrender on every sale

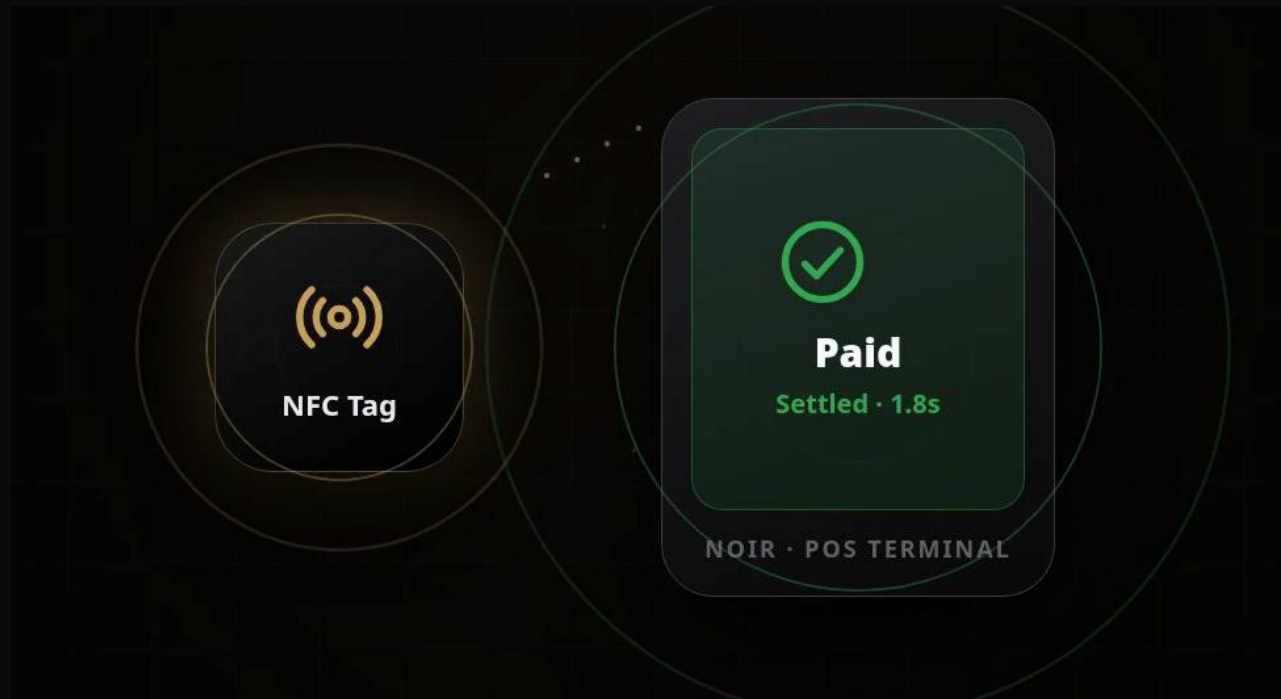
T+2

Days to Settle

Before merchants actually see their money

The principle: crypto gets adopted when people can use it daily. Remove the friction from everyday purchases, and spending crypto becomes as natural as tapping a card — that's what turns holders into users.

Checkout Should Never Be the Bottleneck



What Traditional Payments Demand

- ➔ Unlock your phone and open an app for every purchase
- ➔ Confirm each transaction on-device
- ➔ 2–3% fees on closed card networks
- ➔ Multi-day settlement before funds arrive

Speed should not cost you custody. Even “tap to pay” cards ride closed networks with percentage fees and day-long settlement.

OUR SOLUTION

Introducing Noir Wallet

Zero interaction. Instant settlement.

Tap-First Identity

Any RFID sticker, NFC card, or wearable links to your Stellar wallet — the tag is the login

x402 Protocol

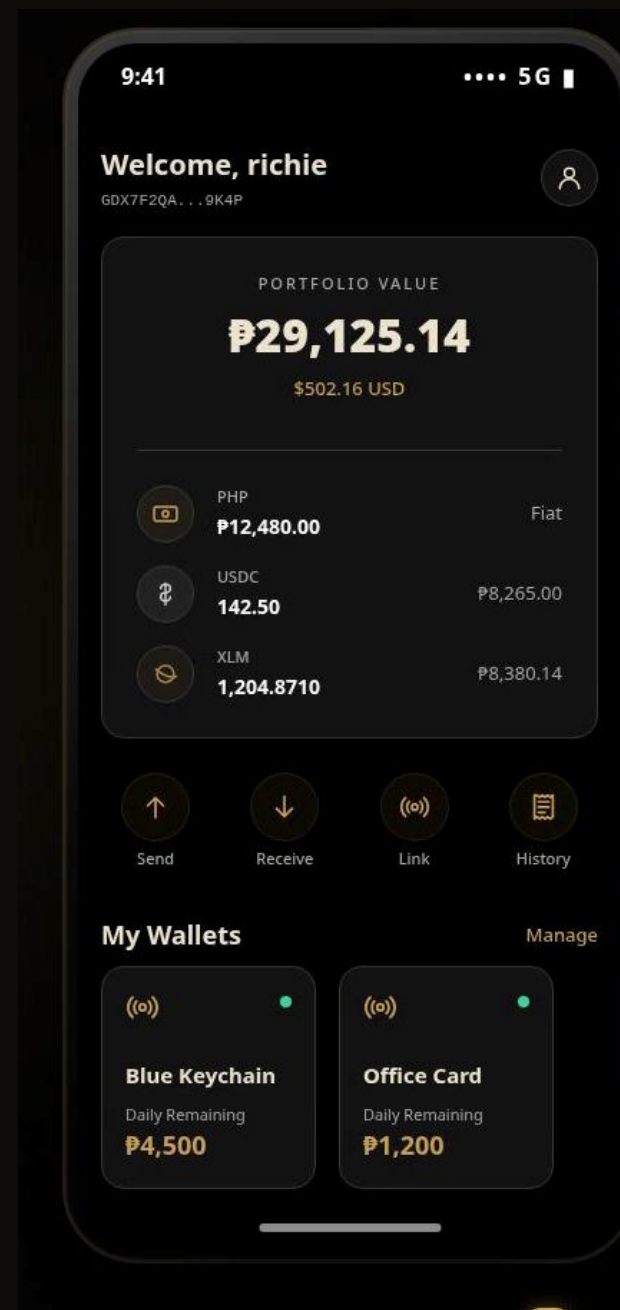
Debited the moment your tag touches the terminal — no unlock, no app, no confirmation

x402 Agents

Each device gets its own funded agent wallet with a daily limit — your main funds stay untouched

Soroban Registry

The device_registry smart contract maps hardware to wallets on-chain — auditable and revocable



Why Stellar?

Stellar provides everything needed for pay-by-tap at machine speed and near-zero cost.

Near-Zero Fees

~\$0.00001 per operation — viable for ₱20 canteen meals

Fast Finality

3–5 second confirmations — faster than a receipt prints

Soroban Contracts

WASM smart contracts for the on-chain device registry

Fee Channels

Sponsored transactions absorb costs — users never buy XLM first

Real Assets

XLM, USDC, and issuer assets — spend and settle in stable value

PDAX Fiat Bridge

Licensed PHP on/off-ramp — merchants cash out in pesos

Architecture Principle

Only hashed device references and payment operations go on-chain. The terminal reads a tag UID, Soroban resolves the linked wallet, and Stellar settles the transfer — with PDAX converting to PHP for merchants who want pesos, not crypto.

~\$0.00001

Per transaction

< 2s

Tap to “Paid”

How It Works

One tap sets off the whole pipeline — the user never sees any of it.

1 · Tap Terminal reads the tag UID over NFC/RFID

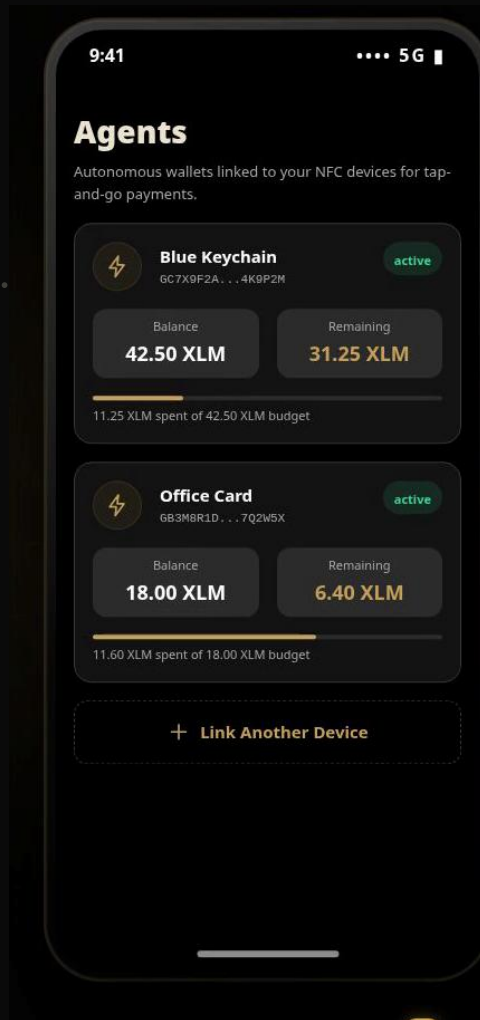
2 · Hash & Resolve UID is SHA-256 hashed; device_registry resolves the wallet

3 · Debit the Agent The device's x402 agent wallet signs within its daily limit

4 · Settle & Cash Out Stellar finalizes in 3–5s; PDAX bridges to PHP

The gateway returns 202 Accepted instantly and settles asynchronously — **under two seconds from tap to “Paid”** — with an offline queue that batches transactions when the network drops.

System Architecture



Three Core Layers

device_registry · Soroban

Hardware-to-wallet mapping with terminal whitelisting and per-device replay protection on-chain

Payment Gateway · Rust + actix-web

Async payment queue, fee channels, daily-limit enforcement, and idempotent transaction records in PostgreSQL

PDAX Fiat Bridge

Licensed Philippine exchange integration — auto-liquidation webhooks convert merchant USDC to PHP

Only SHA-256 device hashes touch the chain. Raw card UIDs never leave the terminal, and no custodial server ever holds user funds.

Security-First Design



Hashed Device Identity

Tag UUIDs are SHA-256 hashed at the terminal — the raw UUID is never stored or transmitted



Terminal Authentication

Only whitelisted POS terminals can initiate payments, via API keys and signed challenges



Replay Protection

Every tap carries a signed nonce; 500ms anti-double-tap rate limiting per device



Blast-Radius Limits

Per-device daily limits and isolated x402 agent wallets — a lost tag never exposes main funds



Encrypted Key Custody

Fee-channel and agent keys sealed with envelope encryption (KMS-backed KeyManager)



Market Opportunity & Competitive Advantage

Target Venues

Transit

Turnstile-speed fares,
no queue pileups

Campus Canteens

Lunch-rush throughput
with student ID stickers

Event Gates

Wristband entry
and cashless concessions

Retail Checkout

Sub-cent fees instead
of 2–3% card rails

₱100B+ QR/contactless volume in PH alone. Every cashless venue that hates checkout queues is a Noir venue.

How We Compare

Feature	EMV Tap	GCash / Maya	Apple Pay	Noir Wallet
No phone unlock needed	✓	—	—	✓
Self-custody of funds	—	—	—	✓
Sub-cent transaction fees	—	—	—	✓
Settlement in seconds	—	—	—	✓
Stickers & wearables	—	—	—	✓
Per-device spend limits	—	—	—	✓
Open smart contracts	—	—	—	✓

Business Model & Roadmap

Revenue Streams

Merchant Transaction Fees

A fraction of card-rail pricing — volume over margin

Device & Tag Sales

Branded NFC cards, stickers, and wearables for venues

PDAX Conversion Spread

Small margin on automated USDC → PHP liquidation

Enterprise & Developer APIs

White-label tap-to-pay for transit operators and campuses

Roadmap

Q3 2026

PDAX production integration, terminal authentication, mainnet pilot with first venues

Q4 2026

Offline queue + batch settlement, merchant dashboard, spending analytics

2027

Transit & campus partnerships, wearable product line, multi-asset auto-conversion

Future

Cross-chain agents, x402 ecosystem APIs, machine-to-machine payments



Tap the future.

Zero-interaction payments, settled on Stellar in under two seconds.

x402 Protocol

Soroban

Stellar

PDAX Bridge

Noir Wallet · Tap. Pay. Done.